

Data Analysis

Build the model

Create design matrix

```
X <- pokemon |>
  select(-c(pokemon, height, weight, type_1, type_2, egg_group_1, egg_group_2,
            generation_id, base_stat_total, hp)) |>
  mutate(intercept = 1,
         dual_type = if_else(dual_type == TRUE, 1, 0),
         dual_type_defense = dual_type * defense) |>
  select(intercept, everything()) |>
  as.matrix()

y <- pokemon |>
  select(hp) |>
  as.matrix()
```

Model

```
fit <- lm(y ~ 0 + X)

tidy(fit) |>
  kable(digits = 3)
```

term	estimate	std.error	statistic	p.value
Xintercept	61.136	3.982	15.354	0.000
Xbase_experience	0.265	0.016	16.671	0.000
Xattack	0.018	0.027	0.684	0.494
Xdefense	-0.188	0.036	-5.201	0.000
Xspecial_attack	-0.074	0.026	-2.885	0.004
Xspecial_defense	-0.048	0.032	-1.507	0.132
Xspeed	-0.234	0.026	-9.062	0.000
Xdual_type	4.090	3.135	1.305	0.192
Xlog_weight	1.934	0.660	2.929	0.003
Xlog_height	7.387	1.543	4.787	0.000
Xdual_type_defense	-0.049	0.040	-1.230	0.219

Our model was selected using domain knowledge about Pokemon and our EDA. We find that base experience, log(weight), and log(height) have positive marginal effects on a Pokemon's HP. We also find that defense, special attack, and speed have negative marginal effects on HP. The remaining predictors, attack, special

defense, dual type, and the interaction between dual type and defense did not have a significant effect on HP at the $\alpha = 0.05$ level. Although outside the scope of our research questions, the model's adjusted R^2 is 0.94, indicating that 94% of the variability in HP can be explained by our predictors.

Even though some of our hypotheses can be immediately tested using our model output, we will provide more explicit testing below.

Recall our hypotheses:

$$H_0^{(1)} : \beta_{\text{speed}} = \beta_{\text{attack}} = 0, \quad H_0^{(2)} : \beta_{\text{dual_type_x_defense}} = 0, \quad H_0^{(3)} : \beta_{\text{base_experience}} = 0$$

Setup helper functions

```
find_beta <- function(y, X) {
  beta_hat <- solve(t(X) %*% X) %*% t(X) %*% y
  return(beta_hat)
}

f_test_quad <- function(y, X, C, d) {
  beta_hat <- find_beta(y, X)

  num <- t(C %*% beta_hat - d) %*% solve(C %*% solve(t(X) %*% X) %*% t(C)) %*%
    (C %*% beta_hat - d)

  den <- nrow(C) * ((t(y - X %*% beta_hat) %*% (y - X %*% beta_hat)) /
    (nrow(X) - ncol(X)))

  f_stat <- num/den

  p_val <- 1 - pf(f_stat, df1 = nrow(C), df2 = nrow(X) - ncol(X))

  return(p_val)
}

make_ci <- function(y, X, beta_hat,
  variance, known_var = TRUE,
  lambda, alpha) {

  if (known_var == TRUE) {
    test_stat <- qnorm(1 - (alpha/2))
  }
  else{
    test_stat <- qt(p = 1 - (alpha/2), df = nrow(X) - ncol(X))
    variance <- (t(y - X %*% beta_hat) %*% (y - X %*% beta_hat)) /
      (nrow(X) - ncol(X))
  }

  lb <- t(lambda) %*% beta_hat - test_stat *
```

```

sqrt(variance * t(lambda) %*% solve(t(X) %*% X) %*% lambda)

ub <- t(lambda) %*% beta_hat + test_stat *
sqrt(variance * t(lambda) %*% solve(t(X) %*% X) %*% lambda)

return(c(lb, ub))
}

beta_hat <- find_beta(y, X) |> as.matrix()

```

Test Hypothesis 1

$$H_0^{(1)} : \beta_{\text{speed}} = \beta_{\text{attack}} = 0$$

We begin with F-tests where we first look at if $\beta_{\text{speed}} = \beta_{\text{attack}}$ and then jointly if $\beta_{\text{speed}} = 0 = \beta_{\text{attack}}$.

```

C_1_equal <- matrix(c(0,0,1,0,0,0,-1,0,0,0,0), nrow = 1, byrow = TRUE)

C_1_zero <- matrix(c(c(0,0,0,0,0,0,1,0,0,0,0),
                    c(0,0,1,0,0,0,0,0,0,0,0)), nrow = 2, byrow = TRUE)

d_1 <- matrix(data = c(0,0))
d <- matrix(0)

pval_equal <- f_test_quad(y, X, C_1_equal, d)
pval_zero <- f_test_quad(y, X, C_1_zero, d_1)

```

Hypothesis	p-value
$\beta_{\text{speed}} = \beta_{\text{attack}}$	0
$\beta_{\text{speed}} = 0 = \beta_{\text{attack}}$	0

Both p-values are nearly 0, meaning that there is evidence at the $\alpha = 0.05$ level that $\beta_{\text{speed}} \neq \beta_{\text{attack}}$ and that either $\beta_{\text{speed}} \neq 0$ or $\beta_{\text{attack}} \neq 0$. To further this, we continue to test whether either are individually equal to 0.

```

C_1a <- matrix(c(0,0,0,0,0,0,1,0,0,0,0), nrow = 1)
C_1b <- matrix(c(0,0,1,0,0,0,0,0,0,0,0), nrow = 1)

pval_speed <- f_test_quad(y, X, C_1a, d)
pval_attack <- f_test_quad(y, X, C_1b, d)

```

Hypothesis	p-value
$\beta_{\text{speed}} = 0$	0
$\beta_{\text{attack}} = 0$	0.494

We now see that there is evidence that $\beta_{\text{speed}} \neq 0$, but we cannot say the same for β_{attack} . To provide some additional insight, we can look at the 95% confidence interval for these coefficients.

```
ci_speed <- make_ci(y, X, beta_hat, known_var = FALSE,
  lambda = c(0,0,0,0,0,0,0,1,0,0,0,0), alpha = 0.05)
ci_attack <- make_ci(y, X, beta_hat, known_var = FALSE,
  lambda = c(0,0,1,0,0,0,0,0,0,0,0,0), alpha = 0.05)
```

The 95% confidence interval for β_{speed} is (-0.285, -0.184). Because 0 is not in this interval, our results align with the F-tests performed above.

The 95% confidence interval for β_{attack} is (-0.034, 0.071). We see that 0 is in this interval, which again aligns with the results seen from the F-tests.

Test Hypothesis 2

$$H_0^{(2)} : \beta_{\text{dual_type_x_defense}} = 0$$

We will perform a single F-test to test this hypothesis.

```
C_2 <- matrix(c(0,0,0,0,0,0,0,0,0,0,1), nrow = 1, byrow = TRUE)
pval_interaction <- f_test_quad(y, X, C_2, d)
```

Hypothesis	p-value
$\beta_{\text{dual_type_x_defense}} = 0$	0.219

The p-value for this F-test is greater than $\alpha = 0.05$, so we fail to reject our null hypothesis. We do not have sufficient evidence that the interaction between dual type and defense has an effect on HP. We can confirm these findings with a 95% confidence interval.

```
ci_interaction <- make_ci(y, X, beta_hat, known_var = FALSE,
  lambda = c(0,0,0,0,0,0,0,0,0,0,1), alpha = 0.05)
```

The 95% confidence interval for $\beta_{\text{dual_type_x_defense}}$ is (-0.127, 0.029). Because 0 is in this interval, our results align with the F-test performed above.

Test Hypothesis 3

$$H_0^{(3)} : \beta_{\text{base_experience}} = 0$$

For our final hypothesis, we will again begin with an F-test.

```
C_3 <- matrix(c(0,1,0,0,0,0,0,0,0,0,0), nrow = 1, byrow = TRUE)
pval_base_exp <- f_test_quad(y, X, C_3, d)
```

Hypothesis	p-value
$\beta_{\text{base_experience}} = 0$	0

The p-value of approximately 0 indicates to us evidence that $\beta_{\text{base_experience}} \neq 0$. A Pokemon's base experience appears to have an effect on its HP. The following 95% confidence interval should confirm this to us.

```
ci_base_exp <- make_ci(y, X, beta_hat, known_var = FALSE,
                      lambda = c(0,1,0,0,0,0,0,0,0,0), alpha = 0.05)
```

As expected, the 95% confidence interval for $\beta_{\text{base_experience}}$ is (0.234, 0.296), which does not contain 0. So, our results align with the F-test performed above.